

TECHNICAL REPORT

Project: 100 MW (AC) / 119.40 MWp (DC) Ground-Mounted Bifacial Solar PV Plant

Location: Eastern Province, Sri Lanka

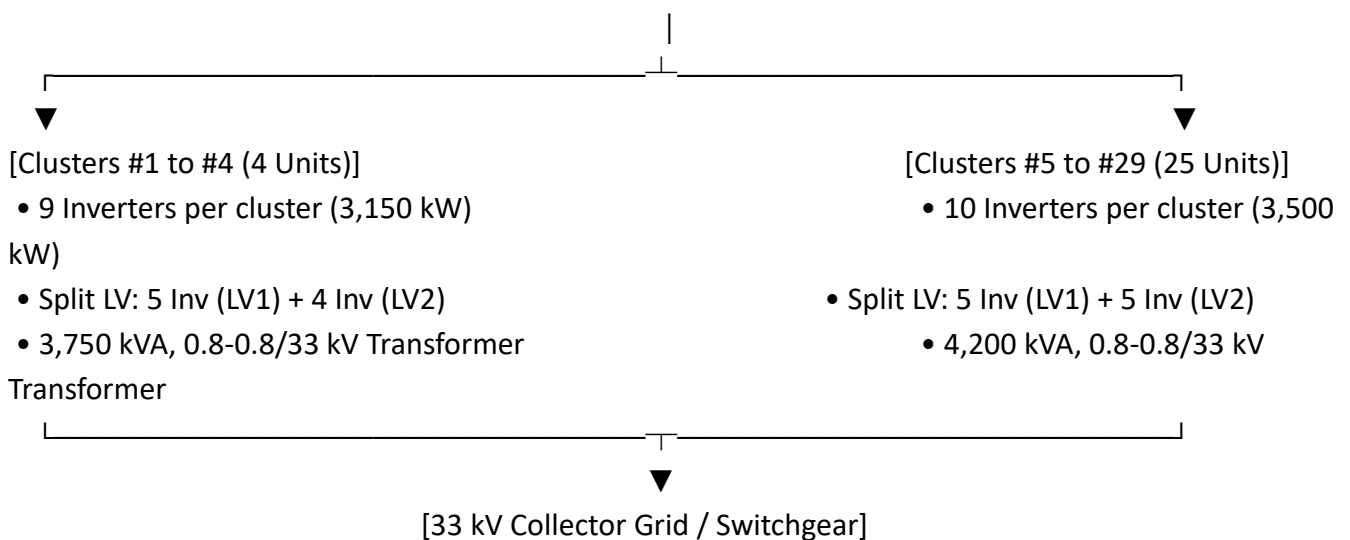
Documents:

- GD-PL-001 (Panel Layout)
- GD-TL-001 (Transformer Layout)
- GD-SLD-001 (SLD, Transformer Units 1–4)
- GD-SLD-002 (SLD, Transformer Units 5–29)

1. Plant System Architecture

Plant Architecture & DC/AC Capacity Verification

[168,168 PV Modules: 710 Wp] ---> [286 Inverters: 350 kW]



Note: Each cluster has 1 Transformer

The plant follows a highly modular, string-inverter-based centralized cluster architecture designed to aggregate at 33 kV for utility export:

- **DC Generation:** 168,168 Bifacial PV modules (710 Wp each) mounted in 2×14 portrait tables at a 7° fixed tilt.
- **String Aggregation:** Modules are wired into strings of 28. Every 21 strings feed directly into a single 350 kW string inverter (no separate DC combiner boxes used).
- **Inversion:** 286 - 350 kW (12-MPPT variant) string inverters convert the 1,500 V *DC* string outputs into 800 V AC (3-phase, 3-wire).
- **LV AC Aggregation:** Inverters are grouped into 29 discrete "clusters." The 800 V AC outputs are gathered via $3C \times 300 \text{ mm}^2$ armored cables into central 3,200 A-frame Air Circuit Breakers (ACBs).
- **MV Step-Up:**
 - Clusters #1 to #4 use 3,750 kVA 0.8/33 kV transformers (9 inverters each).
 - Clusters #5 to #29 use 4,200 kVA (0.8/33 kV) transformers (10 inverters each).
- **Collector Network:** The 33 kV outputs route through MV switchgear to the main 100 MW Collector Substation.

2. System Capacity Verification

2.1 Total PV Module Count

- **Theory:** The total module inventory must perfectly match the product of the array's electrical topology (inverters, strings, and modules).
- **Equation:** $\text{Total Modules} = N_{\text{inverters}} \times N_{\text{strings/inv}} \times N_{\text{modules/string}}$
- **Why this equation:** Physically verifies the bill of materials against the electrical single-line diagram design.
- **Substitution:** $\text{Total Modules} = 286 \times 21 \times 28$
- **Calculate:** 168,168 modules

2.2 Total Plant DC Peak Capacity

- **Theory:** Peak DC capacity is the total number of modules multiplied by the Standard Test Condition (STC) peak wattage.
- **Equation:** $P_{\text{DC}} = \text{Total Modules} \times P_{\text{max,STC}}$
- **Why this equation:** Defines the plant's absolute maximum DC power generation rating for regulatory and economic sizing.
- **Substitution:** $P_{\text{DC}} = 168,168 \times 710 \text{ Wp}$
- **Calculate:** $119,399,280 \text{ Wp} = \mathbf{119.399 \text{ MWp}}$

2.3 Total Plant Nominal AC Capacity

- **Theory:** Nominal AC capacity is the sum of the standard active power output of all string inverters.
- **Equation:** $P_{\text{AC}} = N_{\text{inverters}} \times P_{\text{nom, inv}}$
- **Why this equation:** Establishes the plant's maximum real power export limit to the grid.
- **Substitution:** $P_{\text{AC}} = 286 \times 350 \text{ kW}$
- **Calculate:** $100,100 \text{ kW} = \mathbf{100.10 \text{ MWac}}$

2.4 DC/AC Overplanting Ratio

- **Theory:** Compares the installed DC capacity to the inverter's AC limit. A ratio > 1.0 guarantees the inverters run at full capacity longer during the day despite real-world losses.
- **Equation:** $\text{Ratio} = \frac{P_{\text{DC}}}{P_{\text{AC}}}$
- **Why this equation:** Standard metric used to evaluate inverter clipping vs. levelized cost of energy (LCOE) optimization.
- **Substitution:** $\text{Ratio} = \frac{119.399 \text{ MW}}{100.10 \text{ MW}}$
- **Calculate:** $\mathbf{1.193}$ (Optimal for Sri Lanka's irradiation).

3. PV String Sizing & MPPT Configuration (IEC 62548)

Assumed Environment (Eastern Province): $T_{\text{amb, min}} = 15^{\circ}\text{C}$, $T_{\text{cell, max}} = 70^{\circ}\text{C}$

3.1 Maximum String Open-Circuit Voltage (V_{oc})

- **Theory:** Photovoltaic voltage rises as temperatures drop. The highest voltage occurs at dawn on the coldest day of the year under open-circuit conditions.
- **Equation:** $V_{\text{oc, max}} = N \times V_{\text{oc, STC}} \times \left[1 + \left(\frac{\beta_{\text{voc}}}{100} \right) \times (T_{\text{min}} - 25^{\circ}\text{C}) \right]$
- **Why this equation:** IEC 62548 mandates that this cold-weather maximum voltage must never exceed the system insulation limit (1500 V DC) to prevent fires.
- **Substitution:** $V_{\text{oc, max}} = 28 \times 49.0 \times \left[1 + \left(\frac{-0.24}{100} \right) \times (15 - 25) \right]$
- **Calculate:** 1404.93 V (< 1500 V)

3.2 Minimum String MPPT Voltage (V_{mpp})

- **Theory:** PV voltage drops as the cell heats up. The operating voltage at maximum heat must stay within the inverter's MPPT window.
- **Equation:** $V_{\text{mpp, min}} = N \times V_{\text{mpp, STC}} \times \left[1 + \left(\frac{\gamma_{\text{mp}}}{100} \right) \times (T_{\text{cell, max}} - 25^{\circ}\text{C}) \right]$
- **Why this equation:** Ensures the inverters will not derate or shut down due to under-voltage during hot summer afternoons.
- **Substitution:** $V_{\text{mpp, min}} = 28 \times 40.9 \times \left[1 + \left(\frac{-0.29}{100} \right) \times (70 - 25) \right]$
- **Calculate:** 995.7 V (Fits safely within inverter's 860 – 1300 V window.)

3.3 MPPT Channel Current Loading

- **Theory:** Bifacial modules generate extra current from rear reflections. Paralleling two strings into one MPPT channel sums their currents.
- **Equation:** $I_{\text{mpp, MPPT}} = 2 \times [I_{\text{mpp, STC}} \times (1 + \text{Bifacial Gain})]$
- **Why this equation:** Verifies the current does not exceed the inverter hardware's physical input limit (40 A), which would cause clipping.
- **Substitution:** $I_{\text{mpp, MPPT}} = 2 \times [17.36 \text{ A} \times (1 + 0.10)]$
- **Calculate:** 38.20 A

4. DC Cable Sizing & Verification

4.1 DC Cable Ampacity

- **Theory:** DC strings must handle the maximum possible short-circuit current safely, including bifacial gains and standard regulatory safety margins.
- **Equation:** $I_{\text{design}} = 1.25 \times [I_{\text{sc, STC}} \times (1 + \text{Bifacial Gain})]$
- **Why this equation:** IEC 62548 requires cables and fuses to be rated for a continuous 1.25 multiplier over maximum Isc.
- **Substitution:** $I_{\text{design}} = 1.25 \times [18.40 \text{ A} \times 1.10]$
- **Calculate: 25. 30 A (Compliant.** The drawn 1C x 4 mm² cable carries $\approx 32 \text{ A}$ derated, and the module max fuse is 35 A).

4.2 DC Cable Voltage Drop

- **Theory:** Long cable runs cause resistive voltage drops, losing generated power as heat.
- **Equation:** $\% \Delta V = \frac{I_{\text{mpp, STC}} \times R \times L_{\text{loop}}}{V_{\text{mpp, string}}} \times 100$
- **Why this equation:** Standard ohm's law check to ensure DC losses are kept below the industry standard limit of 2.0%.
- **Substitution:** $\% \Delta V = \frac{17.36 \text{ A} \times 5.88 \Omega/\text{km} \times 0.160 \text{ km}}{1145.2 \text{ V}} \times 100$ (Assuming 80m average run).
- **Calculate: 1. 43% (Compliant, < 2.0%).**

5. LV AC System: Breakers & Cables

5.1 Inverter Output Current

- **Theory:** Apparent power (kVA) dictates the total line current flowing through the AC system at maximum thermal limits.
- **Equation:** $I_{\max} = \frac{S_{\max}}{\sqrt{3} \times V_{LL}}$
- **Why this equation:** Determines the baseline current required to size AC cables, MCCBs, and busbars.
- **Substitution:** $I_{\max} = \frac{352,000 \text{ VA}}{\sqrt{3} \times 800 \text{ V}}$
- **Calculate:** 254.03 A

5.2 MCCB Trip Setting

- **Theory:** Circuit breakers must carry continuous loads without nuisance tripping while protecting cables from sustained overloads.
- **Equation:** $I_{\text{trip}} \geq 1.25 \times I_{\max}$
- **Why this equation:** Code mandates a 25% safety margin over continuous load to account for thermal harmonics and transient fluctuations.
- **Substitution:** $I_{\text{trip}} = 1.25 \times 254.03 \text{ A}$
- **Calculate:** 317.54 A (The drawn **400A MCCB** is **Compliant**).

5.3 LV Busbar and ACB Sizing

- **Theory:** The busbar must safely carry the summed output of all connected inverters.
- **Equation:** $I_{\text{bus}} = N_{\text{inv, per bus}} \times I_{\max}$
- **Why this equation:** Defines the peak current entering the Main Air Circuit Breaker (ACB).
- **Substitution (5-Inverter Bus):** $I_{\text{bus}} = 5 \times 254.03 \text{ A}$
- **Calculate:** 1,270.15 A (The drawn 2000A Busbar and 3,200A ACB are **Compliant**).

6. Transformer Sizing & Grid Margin (IEC 60076)

6.1 Peak Transformer Loading Ratio

- **Theory:** The cluster transformer must handle the combined peak apparent power of its connected inverters without exceeding its rating.
- **Equation:** Loading Ratio (%) = $\left(\frac{N_{inv} \times S_{max,inv}}{S_{transformer}} \right) \times 100$
- **Why this equation:** Ensures the transformer operates safely below 100% capacity to allow for tropical ambient temperature derating.
- **Substitution (4,200 kVA Clusters):** Ratio = $\left(\frac{10 \times 352 \text{ kVA}}{4200 \text{ kVA}} \right) \times 100$
- **Calculate:** 83.81% (Compliant, provides healthy ~ 16% margin).

6.2 Reactive Power Support (CEB Grid Code)

- **Theory:** Transformers must pass enough apparent power to support grid voltage regulation (reactive power) while pushing full real power (active power).
- **Equation:** $S_{req} = \frac{P_{active}}{\text{Power Factor}}$
- **Why this equation:** CEB mandates ± 0.95 power factor support at the Point of Connection.
- **Substitution (4,200 kVA Clusters):** $S_{req} = \frac{3500 \text{ kW}}{0.95}$
- **Calculate:** 3,684 kVA (< 4200 kVA)

7. CRITICAL THERMAL RISK: LV Busduct Analysis (Tray vs Buried)

The SLD specifies 4R x 1C x 630 SQ.MM AL/XLPE/SWA/PVC between the combined LV busbars and the transformer.

- **Required Current:** $\frac{4,200,000}{\sqrt{3} \times 800} = 3,031 \text{ A}$

7.1 Scenario A: Ventilated Cable Tray (In Air)

- **Theory:** Cables grouped on a tray mutually heat each other and are limited by the hot ambient air.
- **Equation:** $I_{\text{bundle}} = N \times (I_{\text{base}} \times C_{\text{ambient}} \times C_{\text{grouping}})$
- **Why this equation:** IEC 60364-5-52 thermal derating calculation to prevent insulation melting.
- **Substitution:** $I_{\text{bundle}} = 4 \times (800 \text{ A} \times 0.91 \times 0.85)$
- **Calculate:** 2,475 A (**FAILS**. Falls short of 3,031 A).

7.2 Scenario B: Buried Duct Bank (In Ground)

- **Theory:** Burying cables inside ducts severely traps heat, causing drastic capacity reductions.
- **Equation:** $I_{\text{bundle}} = N \times (I_{\text{base, soil}} \times C_{\text{soil-temp}} \times C_{\text{duct-grouping}})$
- **Why this equation:** Stricter IEC buried derating standards.
- **Substitution:** $I_{\text{bundle}} = 4 \times (500 \text{ A} \times 0.88 \times 0.65)$
- **Calculate:** 1,144 A (**CATASTROPHIC FAILURE**. Cable insulation will rapidly melt).

Mandated Fix: Discard the 4x Aluminum design. Use **6x Aluminum, 4x Copper**, or a rigid cast-resin busbar trunking system (IP68, 3,200 A).

8. Medium Voltage (33 kV) Fault Withstand

8.1 MV Cable Short-Circuit Adiabatic Check

- **Theory:** During a grid fault, cables must absorb massive thermal energy without the XLPE insulation degrading before the breaker trips.
- **Equation:** $S_{\text{min}} = \frac{I_{\text{sc}} \times \sqrt{t}}{k}$
- **Why this equation:** Determines the absolute minimum cable cross-section (\$\$\$) required to survive the fault.
- **Substitution:** $S_{\text{min}} = \frac{25,000 \text{ A} \times \sqrt{1.0 \text{ s}}}{94}$ (Standard CEB 25kA fault, Al constant 94).
- **Calculate:** 265.95 mm²
- **Verdict:** The drawn 1C x 95 mm² Al cable fails. It can only withstand 8.93 kA and will explode during a severe grid fault. Upgrade to **3C x 185 mm² Cu or 300 mm² Al**.

9. Land Use & Shading Geometry

9.1 Inter-Row Shadow Clearance

- **Theory:** The physical space between PV rows must be longer than the shadow cast by the tables during peak hours (9 AM - 3 PM) at the winter solstice.
- **Equation:** $L_{\text{shadow}} = \frac{L_{\text{table}} \times \sin(\text{tilt})}{\tan(\text{solar altitude})}$
- **Why this equation:** Geometric verification to guarantee no module-level self-shading, which severely limits MPPT yield.
- **Substitution:** $L_{\text{shadow}} = \frac{(2 \times 2.384 \text{ m}) \times \sin(7^\circ)}{\tan(30^\circ)}$ (Assuming portrait mounting and 30° morning sun angle).
- **Calculate:** 1.01 m
- **Verdict:** $1.01 \text{ m} \leq 1.5 \text{ m}$ drawn clearance.

10. Rectification Action Plan (EPC Directives)

1. **33kV Protection Hazard:** Remove the "33kV DO FUSE". Replace with 33kV Vacuum Circuit Breakers (VCBs) or Motorized RMUs linked to transformer alarm trips.
2. **Surge Arresters:** Correct the "Type II" SPD mislabeling to IEC 60099-4 Station Class ZnO Arresters.
3. **Cable Armor:** Change the 1C AC cable armor from SWA (Steel) to AWA (Aluminum) to prevent magnetic overheating.
4. **Cable Sizing:** Upsize the 33kV cable from 95 mm² to 3C x 185 mm² Cu to survive 25kA faults. Redesign the LV busduct (Section 7).
5. **Efficiency:** Downgrade the 4-Pole busbars and ACBs to 3-Pole (3P) to eliminate wasted copper on the 3-wire 800V output.
6. **Data Updates:** Input exact Latitude/Longitude coordinates into the title blocks, and recalculate the 58.52A CT template for the specific 3.75/4.2 MVA loads.